

ORF 245: Fundamentals of Statistics – Spring 2025

Precept 1

Last updated: January 31, 2025

1. (Ice breaking + Introductions)

- Introduce yourself.
- Set communication ground rules: e.g., can stick around for up to 10 minutes after precept. Additional help from office hours.

2. (Set Theory)

(a) Formal and informal ways of describing sets

- $\{2, 4, 6, 8, 10, \dots\}$: The set of all positive even integers.
- $\{\dots, -3, -1, 1, 3, \dots\}$: The set of all odd integers.
- $\{n | n = 2m \text{ for some } m \in \mathbb{N}\}$: The set of all positive even integers
- $\{x | x = 2n \text{ and } x = 3m \text{ for some } n, m \in \mathbb{N}^+\}$: The set of all positive multiples of 6.
- $\{b | b \in \mathbb{Z}, b = b + 1\}$: empty set

(b) Set operations: subset, union $\cup$, intersection $\cap$, complement c . sample space $S = \{x, y, z\}$, $A = \{x, y\}$, $B = \{x, z\}$

- A, B are all subsets of S .
- $A \cup B = B \cup A = \{x, y, z\}$
- $A \cap B = \{x\}$
- $A^c = \{z\}$

(c) Venn diagram: show $A, B, A \cap B, A \cup B, A^c, B \cap A^c$.

3. (Probability Basics Review)

Basic Concepts with example

- *Experiment, Outcome* (e.g. throw a 6-sided die twice)
- *Sample Space S* : the set of all outcomes of an experiment.
(e.g. $\{(1, 1), \dots, (1, 6); (2, 1), \dots, (2, 6); (6, 1), \dots, (6, 6)\}$)
- *Event A* : a collection of outcomes (a subset of S).
- Some example: $A = \{\text{get a sum of at most 10}\}$, $B = \{\text{throw a 1 on the first die OR 5 on the second die}\}$.

Probability

- Three Axioms of Probability
 - (a) (Positivity) For any event A , $\mathbb{P}(A) \geq 0$.
 - (b) (Normalization) $\mathbb{P}(S) = 1$.
 - (c) If $A_1, A_2, \dots$ is a countably infinite collection of disjoint events, then

$$\mathbb{P}\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} \mathbb{P}(A_k)$$

- Intuition: assigned non-negative weights on all events of interest.
- The above example, two different weights: fair die $\mathbb{P}_1$, unfair die $\mathbb{P}_2$ (1/2 probability on 1, 1/10 probability on others).

Note: $\mathbb{P}_1$ and $\mathbb{P}_2$ are two different probability measures.

Properties of Probability

- (a) $\mathbb{P}(\emptyset) = 0$.
- (b) $\mathbb{P}(A) = 1 - \mathbb{P}(A^c)$.
- (c) (Inclusion-Exclusion Principle) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
- (d) (Law of Total Probability) $\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c)$.

Calculation of Probability

- Combine definition with properties.
- (Use (b)): $\mathbb{P}_1(A) = 11/12$, $\mathbb{P}_2(A) = 97/100$.
- (Use (c)) $\mathbb{P}_1(B) = 11/36$, $\mathbb{P}_2(B) = 11/20$.